SIGNIFICANCE

The unmet need. If we are fortunate, each of us will grow old, and reach the point where we, or someone we love, needs help with bathing, dressing, meals, medications, and moving safely through the home. Most families meet that moment at its worst, after a fall or a hospitalization, and are expected to become experts in eldercare overnight (Figure 1).

Figure 1. The fragmented eldercare ecosystem a family must navigate.

The eldercare system in America is fragmented and complex. A family must **find** the right care across the four systems in Figure 1. They must **fund** it: full-time home care costs $80,080 a year,[1] Medicare does not cover custodial home care, and Medicaid does so only in some states, once they have spent down their assets to poverty levels. Public aid programs to help pay for care are difficult to apply for, and an estimated $58 billion goes unclaimed each year.[2] A provider must be able to **staff** it: 63.3 percent of home-care providers reported declining cases in 2023 because they lacked staff.[3] And someone must **execute** the applications, referrals, and intake that stand between a plan and a start date. Many families fall through the cracks and into a vicious cycle of unmet need (Figure 2).

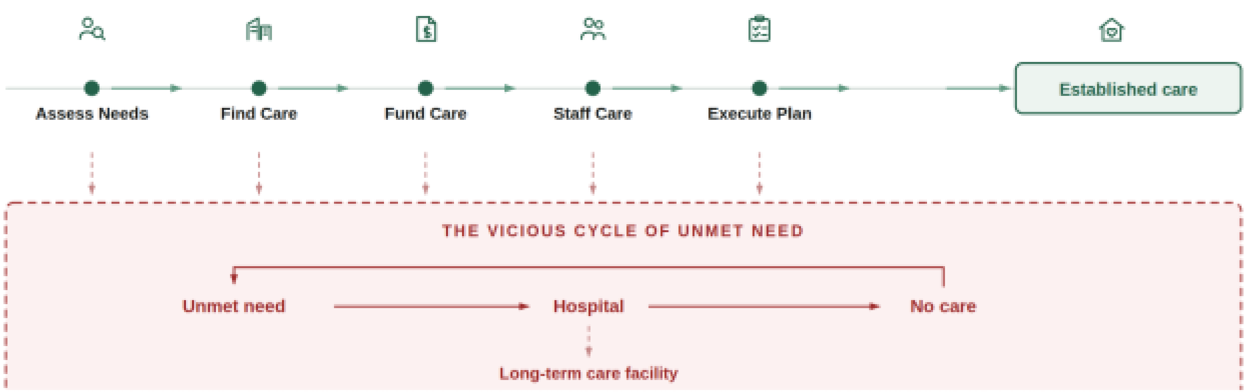

Figure 2. The Care Establishment Pathway and the Vicious Cycle of Unmet Need.

Nearly a third of older adults with difficulty in daily activities go without bathing, meals, or medications in a given month,[4] and unmet needs of this kind independently predict emergency department use, readmission, nursing home placement, and mortality.[5,6,7] Worse yet, unmet daily activity needs are rising because both sources of caregiving support are contracting (Figure 3). The population over 65 reaches 82 million by 2050,[8] family caregivers per adult over 80 fall from more than seven to four,[9] and the *paid caregivers* workforce that would have to absorb the difference faces 9.7 million direct-care job vacancies between 2024 and 2034.[10]

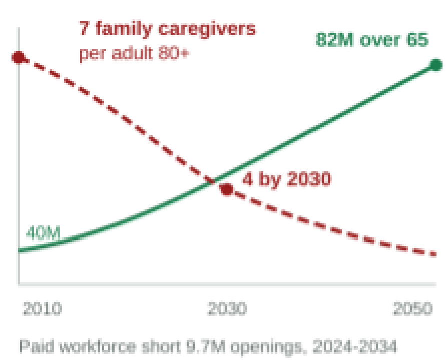

Figure 3. Demand rising while both sources of supply contract.

The product and the north star. Olera helps families find and pay for care, and helps providers find the clients and staff they need to deliver it. Through NIH SBIR Phases I–IIB, we developed CareNavigator, an AI system that assesses a household's needs and means and builds a personalized plan for the care, funding, and providers available to meet them (Figure 4). *But we've found that a plan is not enough.* Families still have to execute the applications, referrals, and intake required to start care, and even a family that successfully navigates those steps may reach a provider without the caregivers to serve them. This CRP addresses both remaining failure points. We will add task-based AI agents that execute and follow up on the administrative steps between a plan and a confirmed care start, and further develop Caregiver Staffing, a product designed to bring new workers into the local caregiving workforce.

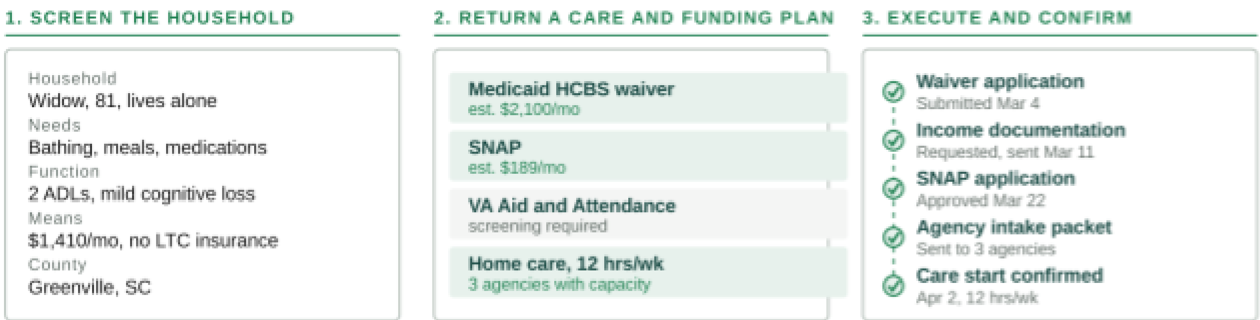

Figure 4. What CareNavigator produces and what the system could then executes and confirm.

Together, these capabilities are designed to complete the care-establishment pathway and ensure capacity exists to deliver it. Our objective with CareNavigator and Caregiver Staffing is to turn more recognized needs into established care before they become unmet needs that feed the vicious cycle.

Market segments and customers. Families use CareNavigator for free, and providers participate in the navigation network for free, so payment does not gate the care-establishment pathway (Table 1). Olera instead monetizes two additional sources of value through two paying customer classes: providers and risk-bearing institutions, including insurance payers and accountable care organizations (ACOs).

Segment	Olera value	Relationship	Why it is marketable
Families	CareNavigator: navigation through the care-establishment pathway	Free user	Free access keeps payment from gating care establishment.
Care providers	CareNavigator participation and family connections	Free participant	Free participation preserves provider access without a paid referral gate.
Care providers	Caregiver Staffing: recruit and place new caregivers	Paid customer	Staffing is a major provider need; providers already pay to solve it, and capacity is required to establish care.
Insurance payers and ACOs ("risk-bearers")	CareNavigator for defined member or patient populations	Paid customer, evidence-gated	They bear downstream costs when care is not established; navigation can create value by establishing care earlier.

Table 1. Olera's market segments, commercial relationships, and sources of value.

Caregiver Staffing is the near-term commercial beachhead. Staffing is a major problem for healthcare and long-term services and supports providers, and they already spend substantially to recruit and replace caregivers. It is also the half of the problem families cannot solve on their own. A family that finds care and can pay for it still has nowhere to go if the provider has no one to send.Because care cannot be established without provider capacity to deliver it. Caregiver Staffing therefore turns a problem providers already pay to solve into a marketable product that also strengthens CareNavigator's care-establishment infrastructure.

CareNavigator creates a second market. Risk-bearing institutions, including Medicare Advantage plans, ACOs, and Medicaid managed-care organizations, bear downstream costs of hospitalizations and institutionalization of older adults. These organizations can pay for CareNavigator to establish care for defined member populations. Medicare's GUIDE model provides early precedent for payment for care navigation and caregiver support. However, the market remains evidence-gated, and therefore our CRP must generate defensible evidence that CareNavigator can increase care establishment and create economic value for these buyers.

Competitive environment and our advantage. The competitive environment mirrors the fragmented care-establishment pathway described above. Existing human and technology-enabled alternatives address individual steps or substantial portions of it, but generally operate through separate services and business models (Table 2). That fragmentation matters because responsibility can return to families between services, creating opportunities for care to fall through before it is established, which then feeds the same vicious cycle.

Competitive category	Representative examples	Assess	Find	Fund	Execute	Establish	New workforce	Open access
Public information and benefits	Eldercare Locator; BenefitsCheckUp	•	●	●	•	○	○	●
General-purpose AI	ChatGPT; Claude; Gemini	●	●	●	•	○	○	●
Human navigation	Discharge planners; case managers; private care managers	●	●	●	●	● / •	○	•
Digital and hybrid navigation	Wellthy; Homethrive; Cariloop	●	●	●	●	● / •	○	•
Senior-care referral marketplaces	A Place for Mom; Caring.com	•	●	○	•	•	○	•
Workforce recruitment	Indeed; myCNAjobs; staffing agencies; referrals	○	○	○	○	○	•	○
Olera after the CRP	CareNavigator with Caregiver Staffing	●	●	●	●	●	●	●

Table 2. Competitive alternatives across the care-establishment pathway. ● core capability · • partial or variable · ○ not typical.

Olera engineers from first principles to solve for the endpoint that matters–established care. This is why CareNavigator was designed to instrument the full care-establishment pathway so we can measure where, why, and how quickly care establishment succeeds or fails, and target interventions to the failure points. Four architectural advantages follow from this first-principles approach:

1. Digital scale. The care establishment pathway can be screened and navigated across populations through the open web and mobile application architecture.

2. Open to all. Families can enter without payment or institutional sponsorship, and providers can participate without paying for referrals. This preserves broad, neutral participation while giving the instrumented pathway visibility across the most families and providers.

3. Qualified demand and capacity. CareNavigator can connect providers at no referral cost with families whose needs, funding options, and service fit have been characterized. Caregiver Staffing can mobilize new qualified labor pools rather than redistribute existing workers.

4. Instrumentation and evidence. CareNavigator follows the case to the meaningful endpoint of established care rather than ending wherever an individual service's function ends. This creates evidence about whether the care establishment pathway worked, where it failed, and how long it took at the county level.

The CRP tests whether these proposed advantages are real, measurable, and commercially meaningful.

Related development efforts in academia and industry. Existing evidence spans care coordination, home-based care, digital caregiver support, closed-loop referral, and workforce development, but these efforts have largely developed separately. Care-coordination research has shown the value of structured support, including increased use of home- and community-based services and dementia outpatient care. Home-based interventions such as CAPABLE have reduced disability and supported aging in place, with Medicaid analyses also finding lower monthly spending. Digital caregiver interventions have shown favorable effects on caregiver health, self-efficacy, skills, quality of life, support, and coping, although application-based interventions show variable engagement and effectiveness and many remain prototypes. Our family-engaged research has verified the usability and acceptance of the built platform functions and evolving needs of families to have a one-stop shop of care services that can establish the care needed. Closed-loop referral research identifies integration, continuity, and outcome measurement as unresolved priorities. Caregiving workforce research shows that retention depends on wages, training, job quality, and working conditions, while tuition assistance, career development, recognition, leadership, and career pathways are among the strategies being studied to strengthen the workforce. Together, these literatures and the industry environment discussed previously support the individual components of Olera's approach while exposing the need for an end-to-end solution.

Hurdles to adoption. A coherent product still fails if the people required to use it do not adopt it. Families must trust CareNavigator enough to move from receiving information to allowing AI-enabled execution of consequential care-related tasks. Providers must receive qualified families and workers with sufficient value and minimal workflow burden to incorporate CareNavigator and Caregiver Staffing into their operations. Caregiver Staffing must create genuinely new, safe, reliable workforce supply that providers will hire and that workers will remain in long enough to create real capacity. Risk-bearing institutions face the strongest evidence threshold. Their adoption depends on demonstrating that CareNavigator is accepted by their members and that the care establishment it produces creates defensible economic value through utilization reduction. The CRP project is therefore designed to address these specific adoption and commercialization risks.

INNOVATION

Key Innovation 1: making the care-establishment pathway computable. Phase IIB modeled the front end of eldercare planning: household needs and means, likely benefits and aid, and appropriate services and providers. The CRP addresses the harder engineering problem that follows. Eldercare unfolds through a finite but highly variable set of entities, documents, communications, decisions, and delays that differ by household, program, provider, and geography, and reliable automation first requires a computational representation of that system.

We propose a longitudinal Care Establishment Model organized around seven eldercare-specific domains, each carrying substates developed in Aim 1 (Figure 5). The Phase IIB eldercare LLM interprets each of these inputs and normalizes them into this common state. The model can then assemble a household's verified state into the program- or provider-specific information a computational workflow, like an AI agent, requires. Applications, documentation requests, provider denials, waiting lists, service starts, and disruptions can then become timestamped events with geography and provenance in a normalized database rather than lost in disconnected inboxes and records or scattered elsewhere along the care establishment pathway.

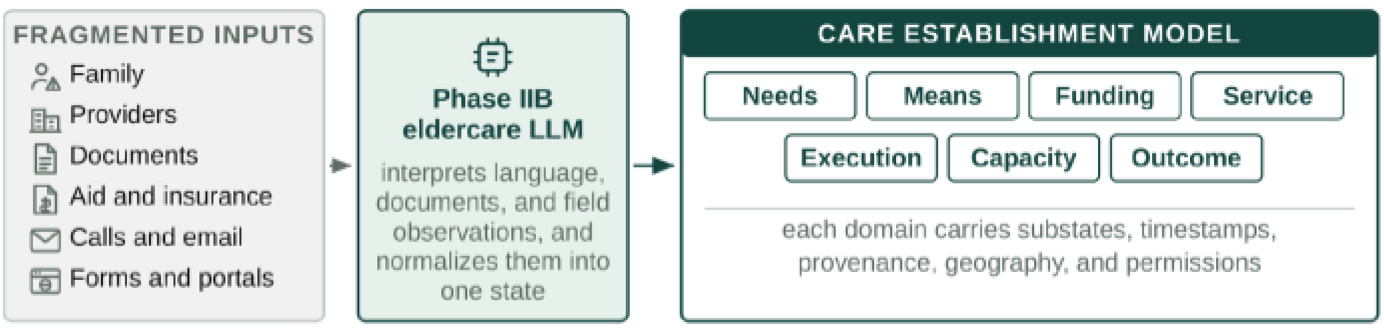

Figure 5. Fragmented eldercare inputs are normalized into one longitudinal Care Establishment Model.

This is the load-bearing advance for the rest of the CRP. Software cannot reliably automate a fragmented pathway unless it can observe what state a case is in, what changed, who owns the next action, and what evidence defines completion. It also makes the pathway measurable. Executed cases reveal precisely what happened between recognized need and established care, allowing us to observe where, why, and how quickly care establishment fails at the household and geographic levels. To our knowledge, there are currently no established models of the eldercare ecosystem we can apply directly to our use case of helping families navigate the eldercare ecosystem families find themselves navigating as described in Figure 1.

Key Innovation 2: AI agents that execute and learn from the care-establishment pathway. The prevailing paradigm is that digital tools inform, recommend, refer, or plan, and families execute tasks to secure care. The CRP explores offloading the latter to reduce failure points in the care-establishment pathway by adding specialized AI agents that perform administrative tasks to secure eldercare aid and services (Figure 6).

AI agents combined with LLM reasoning with the care model described in Innovation 1, could act on the household’s behalf, bounded by explicit workflow logic, deterministic permission gates, and constrained tools to execute key steps to securing care. Key technologies incorporated into the agent system can include APIs, permissioned browser automation where portals require direct interaction, document handling, email, SMS, fax, or appointment scheduling. For example, a user may authorize the system to prepare an LTSS application, contact multiple home-care agencies with the same care details, schedule transportation, or utilize insurance benefits. Actions requiring attestation, legal authority, or consequential family choice can be identified and bounded so they require approval and remain human-controlled.

Episodic AI agent use is helpful, but the care establishment pathway may require multiple tasks over time. This is an engineering challenge that requires AI to act with persistence across delays in intake processing, incomplete information, and heterogeneous external systems. Execution must then follow an event-driven loop capable of interpreting a plan, outlining tasks to be executed now or queued for later, waiting or returning, and verifying or escalating items over longer periods of time as cases should continue until the endpoint of care establishment is verified, declined by the family, or cannot safely proceed.

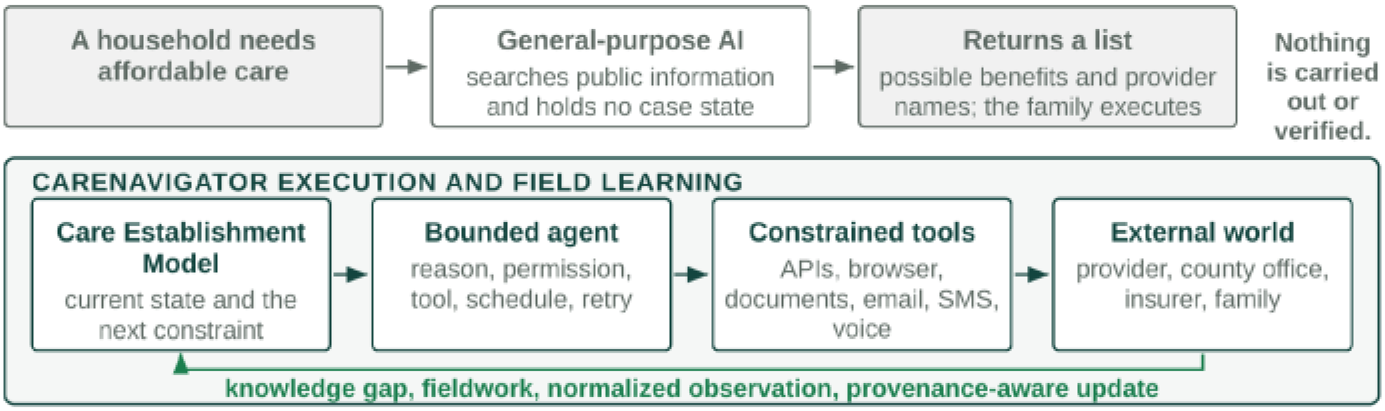

Figure 6. General-purpose AI returns information. CareNavigator executes, observes, and writes back.

The second novelty of eldercare AI agents acting along this model is that execution itself generates field knowledge. When the model meets an unresolved operational question or gap in data substrate, the same tools can retrieve current information, inspect a portal, contact a representative, or escalate fieldwork to a person, and the LLM normalizes what comes back and stores the findings with source, geography, time, and provenance. A provider may publicly accept a payer yet report by phone that weekend capacity is unavailable until November. That fact retrieved by the agent and stored in the model, improves the current case and every subsequent case in the same market. More executed cases therefore expose more knowledge gaps, produce more direct observations, and can improve future effectiveness and data integrity, which matters significantly in eldercare because operational truth is local, rapidly changing, and largely absent from the public web.

Key Innovation 3: data-directed creation of new caregiver capacity and eldercare interventions. Better information and execution still cannot establish care when no person is available to deliver it. Long-term care capacity ultimately depends on the caregiver workforce, and existing staffing channels predominantly compete

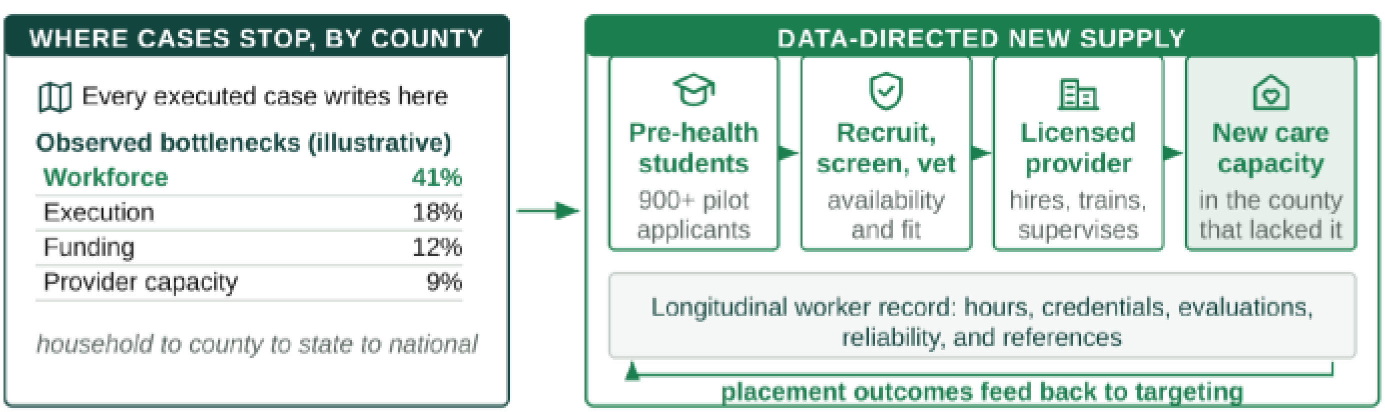

Figure 7. Instrumented cases show where capacity fails and direct new supply at that deficit

for workers already circulating in the same constrained labor market. Olera instead uses the instrumented pathway to identify where workforce is actually preventing care, and targets new supply at those deficits (Figure 7).

CareNavigator can combine funded family demand, provider denials and vacancies, service type, geography, shift requirements, worker availability, placement outcomes, and external workforce indicators into a local capacity view. To fill workforce vacancies, we are exploring a specialized new caregiver entrance pathway for pre-health students who are uniquely well suited for caregiving roles because they are distributed through universities, a new cohort becomes available each year, often seeking documented patient-facing experience, and available for the evenings and weekends providers report are hardest to fill. An existing pilot of this concept generated more than 900 undergraduate applicants, showing that this is a reachable labor pool. Further, we are designing a longitudinal worker record into our model that accumulates verified hours, experience, credentials, provider evaluations, reliability, and references to further enhance visibility into the eldercare system on the individual care worker level.

The same instrumentation has value beyond staffing. Households' states aggregate into county, state, and national visibility on unmet needs, funding failures, processing delays, provider deserts, workforce shortages, and time to established care to create an analytic layer for researchers, industry, risk-bearing institutions, and public agencies. Caregiver Staffing is the first intervention we are exploring to target local markets based on these analytics because the workforce shortage is already large, and caregiver shortages are a major fall-off point along the care establishment pathway.

The end product: the Olera CareNavigator. The three innovations converge behind the existing CareNavigator web and mobile experience, which Phase IIB developed through repeated build-measure-learn cycles with family caregivers. The front end stays intentionally simple while the computational and agentic infrastructure works behind it. A hospital discharge, a family that cannot afford home care, and a provider without staff enter by different routes and run through the same infrastructure to the same place (Figure 8).

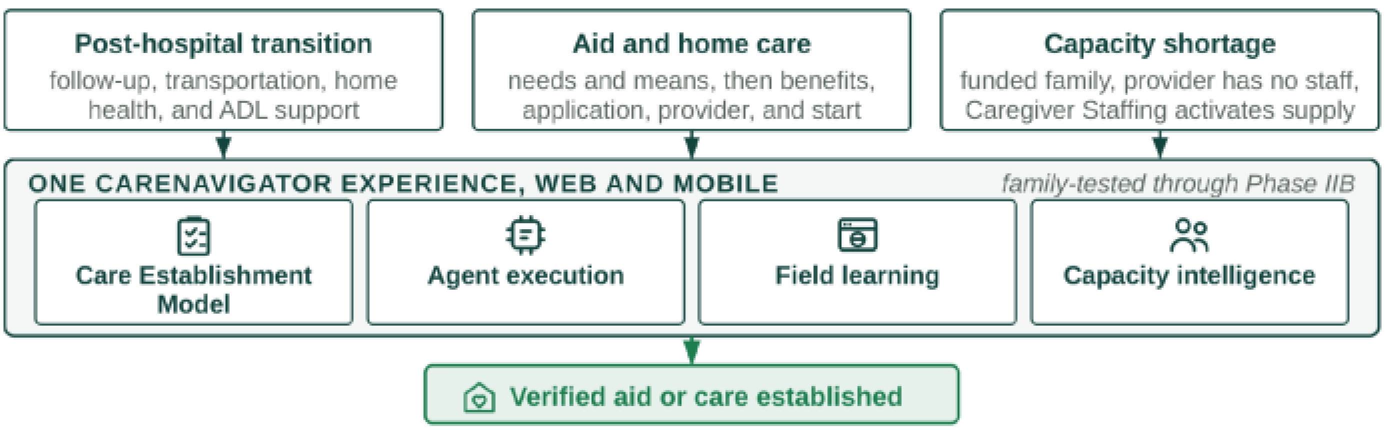

Figure 8. One CareNavigator experience, different paths, one verified endpoint.

APPROACH

Specific Aim 1: Engineer and technically verify the integrated care-establishment infrastructure. Aim 1 builds on R&D from Phase I-IIB where we established CareNavigator's web and mobile experience, eldercare-tuned LLM, national provider and benefits data infrastructure, household needs-and-means modeling, and eligibility and service matching. The tasks described below add a longitudinal Care Establishment Data Model, eldercare AI agent workflows, and Caregiver Staffing infrastructure required to facilitate and observe the full care establishment pathway for users.

Task 1.1: Build the Care Establishment Pathway Model. We will formalize the computational state architecture introduced in Innovation 1 around seven eldercare-specific domains we have identified from prior research to span the care establishment pathway. At a high level, these include normalizing input from families, aid programs, and service providers by needs, means, funding, service, execution, capacity, and outcome. Each domain will contain substates including timestamped events, geography, provenance, evidence, ownership of the next action, and explicit terminal states. The Phase IIB eldercare LLM will serve to normalize natural language input from family text or voice dialogue, uploaded documents, forms, and provider or program in-kind input and intake requirements into this common data representation. Specific programs or providers will span public aid, insurance benefits, home care, assisted living, home health, transportation, and post-discharge coordination and will be encoded and audited for whether the model scheme accommodates the data required to determine the next administrative action a case may take towards establishing aid or care.

Task 1.2: Build a suite of eldercare AI administration agents. The Care Establishment Model will be coupled to and trigger both LLM reasoning and eldercare-specific AI agents with explicit workflow logic, deterministic permission gates, scheduling, and constrained tools. Creation of these agents will incorporate best practices for safety and privacy as defined by IBM AI Agent Security Best Practice guidelines. Family or provider input into the model through the LLM will be normalized and mapped to a suite of AI agents whose tools and workflow parameters are designed to execute the functions needed to complete administrative tasks. Tools include APIs to software for permissioned browser automation for portals, document assembly software, email clients, SMS, fax, phone calls, and meeting scheduling. Each agent workflow will be an event-driven loop triggered by mapped model input and output actions such as obtaining permissions from the user, acting on their behalf, waiting or returning to act at a specified time, verifying the terminal state of a defined task or escalating for user or support team manual processing. Actions requiring legal authority, attestation, identity verification, or clinical judgment will be gaurdrailed so execution does not cross medicolegal boundaries or enact clinical activity like diagnosis, treatment, or interpretation of clinical information. Additional safeguards include permissions that distinguish the person using the platform from the person whose information or benefits are involved, and agents will not be allowed to infer legal decision-making authority from caregiver status alone. Unsupported, stalled, or ambiguous workflows will be escalated to the user or our human support team for manual processing. The internal engineering team will test agents on case simulations in an internal staging environment to verify intended administration tasks along the eldercare establishment pathway are functioning as intended. The result is a suite of eldercare administrative AI agents capable of the key execution functions required when interpreting eldercare-related input from users, interacting with tools to execute required tasks related to care establishment, and following up over time to confirm completion of necessary intake tasks.

Task 1.3: Build a field learning closed-loop data architecture. Execution agents will also function autonomously as structured field researchers. If an agent working a case encounters missing, stale, or conflicting operational information in the model or database, the system can autonomously retrieve and store the missing data. Specifically, agents will be allowed to search current public web information, inspect an available application portal, contact the relevant organization through an approved communication channel like email, or escalate the question to our support team. Returned information will be normalized with source, geography, time, and provenance and will enter the production knowledge layer only after the required quality check. Time-sensitive facts such as provider capacity, office representatives, accepted payment sources, waiting-list status, and specific application procedures will be preserved for provenance and to improve the learning of the system so it delivers more effective execution functions over time. This functionality will be tested on similar mock cases to Task 1.2, and we will observe that autonomously collected data is normalized into the model without error, and that new data entries trigger a human quality assurance check before persisting in the model for the next case set.

Task 1.4: Develop Caregiver Staffing infrastructure. Olera's existing student recruitment and provider-facing placement infrastructure will be extended from applicant intake to verified employment. Candidates will be screened for availability and fit based on individual providers' specifications stored in the model and verified autonomously by decision tree logic before provider handoff. Licensed providers retain responsibility for hiring, training, employment, supervision, and any provider-specific screening. After hand-off, employer-confirmed hours, experience, populations served, evaluations, reliability, and references will accumulate into a longitudinal worker record that is stored for geographic workforce capacity modeling.

Task 1.5: Preliminary usability and acceptance testing before real-world deployment. Under Clemson University IRB, we will conduct a mixed-method stakeholder study of the usability of the interfaces and agent execution relevant to the three user classes, including family caregivers (n=25), service providers (n=25; non-medical home care operators), and potential student workers (n=25) who will complete moderated sessions using standardized non-live cases simulating the functions relevant to their roles on our non-production staging environment. Study participants will be recruited from Olera's existing pipeline funnels for each class, including paid advertisements, business-line communications to providers, and existing and newly formed university relationships with pre-health advising offices. During sessions conducted on video call with screen-sharing functionality, participants will interact with the relevant interfaces for their user class and think aloud while completing role-specific tasks. We will observe behavior and record task success, time, errors, and assists, administer the System Usability Scale (SUS) for all participants, and the Trust in Automation Scale (TIAS) for families who use AI agent execution functions. Semi-structured interviews post-session will focus on workflow comprehension, permission boundaries, trust, and needed interface or software functionality refinements. This task therefore establishes whether intended family users find it useful and trust and accept the permission architecture under controlled conditions before Aim 2 real-world testing.

Potential problems, alternatives, and GO/NO-GO to Aim 2. Failed or malfunctioning AI agent workflows can be narrowed, disabled, moved behind additional approval, or thresholds lowered to trigger human manual processing while it is corrected, reverified, or retired. Progression of the integrated data model, AI agent suite, and caregiver infrastructure to Aim 2 real-world testing is a NO-GO if any critical workflow class has an unresolved critical event defined that would break best practice in AI safety and privacy, unresolved material permission failure, systematic misunderstanding of consequential-action boundaries, or inability to enforce the verified execution scope. Specifically, Aim 2 is gated only after the integrated system meets the prespecified technical criteria in Table 1 below:

Measure/Deliverable	Criterion	Source
Care Establishment Data Model	Complete computational model across pathway steps including needs, means, funding, service, execution steps, workforce capacity, and care establishment endpoint outcome	Internal schema audit; Task 1.1
Agent workflow execution	All expected actions per internal team assessment with no perceived errors in workflow logic, permission gating, tool constraints, or agent output	Technical test set with mock cases in simulated internal staging environments; Task 1.2
Agent field learning with closed-loop provenance and QA	Operational data gaps in our database are filled by agents autonomously, and notifications for QA by human team; 100% of new data acquired, normalized, and persisted in the model	Technical test set with mock cases in simulated internal staging environments; Task 1.3
System Usability Scale	Mean ≥72 for each participant class	Moderated sessions; Task 1.5
Trust/permission comprehension	Mean TIAS ≥5 and no critical control failure	Moderated sessions; Task 1.5
AI Agent Security Best Practice	No unresolved critical event breaking best practice guidelines for safety and privacy	IBM AI Agent Security Best Practice; GATE

Table 1. Aim 1 success criteria and the source of each measurement.

Specific Aim 2: Test whether the integrated system establishes aid and care and creates value in real markets. The verified suite of Eldercare AI Agents and the Caregiver Staffing infrastructure developed in Aim I will be validated in real-world operations without pay gates (i.e., free pilots). The primary question is whether a critical number of tracked family episodes reach the confirmed aid or care establishment state or, if not, why pathways succeed, stall, or fail. We will also observe the Caregiver Staffing infrastructure in action and observe if it is successful in creating new care staff capacity with providers experiencing staffing vacancies.

Task 2.1: Validate CareNavigator with families in real market environments. Aim 2 will track approximately 440 family navigation episodes across eight medium-to-large-sized counties (i.e., "markets") over nine months. Markets will be selected using specified criteria for family demand, provider density, relevant benefit and service availability. Families will be recruited via Olera's organic web activity, social media accounts, and paid advertising targeted to the market radius. Families will use a self-service portal on the web or mobile app platform interface. Over the nine-month observation period, the instrumented system components will record family navigation episodes and persist to the model state transitions (i.e., assessed, in-progress, denied, accepted, etc.), task execution, communications, escalations, and field-learning events. Each action will also be classified as user-executed, agent-executed, or Olera support team-executed. Each episode may be composed of a variety of required tasks assigned to various agents to complete towards the specific terminal success state of care established. The establishment of the care success state is assigned during the assessment stage for each episode and followed to completion or termination. Completion requires family confirmation and, when possible, provider or aid program representative confirmation that the documented primary need was actually obtained. This confirmation can be ascertained by the system through communication from users through the platform interface (i.e., provider or family logging successful establishment), or through agent-initiated communication using email, SMS, or phone. The primary success outcome is care established per episodes initiated. Secondary outcomes include whether the agentic infrastructure, rather than a human (either user or Olear support team member), performed the administrative work, time to establishment, specific task failure or user fall-off point, and reasons for task delay or failure ascertained by user error reports. Among families offered a permission-gated agent action, we will also measure rates at which they authorize or deny.

Task 2.2. Validate Caregiver Staffing in free pilots with non-medical home care providers. Caregiver Staffing will be validated in the same eight markets as Task 2.1 to increase workforce capacity where care navigation is aggregating provider demand. The decision to make the pilot free to providers is to increase participation and therefore throughput so we can learn from operational experience and see system-wide what works and what could be improved. Over the nine months, we will target 15 non-medical home care agencies and 150 pre-health student worker placements. Non-medical home care providers will be recruited through paid advertising and business line calls offering the product for free as a supplemental hiring pipeline. Student workers will be recruited through local university career advising centers and student organizations. All will enter through the self-serve platform that hosts provider and student profiles, an interview scheduling and hiring system, and a longitudinal worker log and record system. Provider staffing requests, applicants referred, successful hires, rejections, first shifts, total hours worked, employment duration, and termination of employment will be monitored through the instrumented platform components.

Task 2.3: Conduct post-use mixed-methods evaluation and pre-commercial learning. After exposure to the platform-service, users of both CareNavigator and Caregiver Staffing will be invited to provide feedback on the experience and value added via surveys and feedback. For families who participated in navigation episodes, standardized surveys will measure acceptability, appropriateness, feasibility, perceived value, workflow fit, user experience, and comfort delegating different categories of administrative action. Interviews will specifically examine which actions users authorized, declined, or preferred to keep human-assisted, why those boundaries differed, and whether agent execution reduced or introduced administrative burden. For providers, surveys and interviews will focus on determining provider willingness to pay for the care workers they hired, and this price will be treated as a pricing hypothesis for Aim 3. For student care workers hired, we will focus feedback on their experience with platform onboarding and vetting, employer relationships, client hours and experience, and value to their future healthcare career goals. Interview transcripts will be analyzed using a shared codebook. The resulting qualitative and quantitative user data will set the stage for Aim 3 experimentation in eight new markets, specifically setting the price to start with for Caregivers Staffing.

Analysis, alternatives, and GO/NO-GO. Aim 2 outcomes will be analyzed across all eight markets and reported by market where informative. For CareNavigator, we will estimate the proportion of family episodes reaching verified care or aid establishment and the proportion of eligible administrative actions completed by AI agents without human execution. For Caregiver Staffing, we will measure the full recruitment-to-employment pathway, including placement yield, hires per provider, hours worked, and employment duration. Acquisition and operating costs will be measured across families, providers, and workers. If individual agent workflows or Staffing processes underperform, we will identify the failure point, refine the workflow, and retain human support where automation is not yet reliable. For staffing, the willingness to pay per-hire price will be obtained via the qualitative and quantitative feedback and used for real pricing in Aim 3. Go/No-go decisions to progress CareNavigator or Caregiver Staffing to Aim 3 will be based on meeting the following criterion:

Measure	Criterion	Source
Family episodes enrolled	~440 enrolled to retain ≥ 400 analyzable episodes after ≤ 10% withdrawal/unusable outcomes data	Enrollment records; Task 2.1
Verified care or aid establishment	GO if lower bound of 95% CI >35% of episodes reach aid or service specified by agent's assessment primary need	Family confirmation + source records; Task 2.3
Family authorization of consequential agent action	≥60% of eligible families offered such an action authorize ≥1	Permission and execution logs; Tasks 2.2–2.3
Agent-executed administration actions	≥70% of approved agent actions eligible for automation completed without human execution	Platform action logs + navigator records; Tasks 2.2–2.3
Workforce placement yield	≥25% of qualified workers entering provider placement reach employer-confirmed hire	Worker + employer records; Tasks 2.2–2.3

Provider staff connections quality and quantity	Number of staff hires per provider, hours worked, duration of employment	Platform + provider/worker records
Provider willingness to pay	Free pilot completers will assess the service's value, provide a qualitative assessment of their willingness to pay, and estimate the price they would be willing to pay per successful hire.	Feedback form provider free pilot users; Task 2.2
CAC and cost to serve	Measured across all three stakeholder groups, including acquisition costs, platform usage costs, and all associated fixed and variable operating costs.	Cost ledger; Tasks 2.1-2.4
AI Agent Security Best Practice	No unresolved critical event breaking best practice guidelines for safety and privacy	IBM AI Agent Security Best Practice; GATE

Table 2. Aim 2 success criteria, and the source of each measurement.

Specific Aim 3: Test whether demonstrated value supports repeatable paid commercial economics.

Aim 3 will test whether the operating model validated in Aim 2 can be reproduced in new markets under commercial conditions. We will enter eight new markets, acquire families, providers, and workers through the channels established in Aim 2, and operate both CareNavigator and Caregiver Staffing. CareNavigator will remain free to families and providers while Caregiver Staffing will transition to a paid provider product. CareNavigators ability to establish care will be tracked and used to model its potential value to payers and ACOs in preparation for a post-CRP proof of concept.

Task 3.1: Replicate the integrated system and test paid economics in eight new markets. Aim 2 will provide direct evidence on provider value of the Staffing program, hires completed, worker retention, willingness to pay, and Olera's cost to operate. These findings will determine the Caregiver Staffing initial price tested in Aim 3. Over another nine-month period, we will enter eight markets not used in Aim 2. In each market, Olera will recruit new families into CareNavigator, activate local providers, and recruit student workers using the acquisition channels validated in Aim 2. CareNavigator family episodes will again be followed from assessment through verified care or aid establishment using the same core outcome definition established in Aim 2, allowing us to determine whether care-establishment performance replicates in previously untested markets. Caregiver Staffing will operate concurrently, but providers will now be offered the paid product at the price informed and set by willingness to pay information in Aim2. We will measure family episodes and verified care establishment, providers and workers acquired, staffing requests, hires, and retention, paid conversion and repeat purchasing. Revenue, customer acquisition cost and cost to serve be used to calculate service margin.

Task 3.2: Translate the CRP evidence into the post-CRP commercialization plan. At the nine-month mark, we will begin to integrate evidence from Aims 2 and 3 for the CareNavigator towards a post-CRP evidence package that will show institutional buyers like Medicare Advantage and ACOs the ability for the CareNavigator to establish care. Published evidence and external economic modeling may be used to estimate the potential downstream value of improved care establishment. These results will define the target population, study design, outcomes, data requirements, sample size, and follow-up for a post-CRP institutional proof-of-concept testing whether improved care establishment reduces utilization or total cost. For Caregiver Staffing, measured purchasing, retention, acquisition cost, cost to serve, contribution margin, and replication across markets will replace the assumptions in Olera's commercial model and determine whether the provider-paid product is ready for broader expansion.

Potential problems, alternatives, and final commercial GO/NO-GO. Commercialization development of the CareNavigator is a go if it is shown to establish care replicable in another 8 markets in Aim 3 and economic modeling of potential reduced costs are promising. Caregiver Staffing is a GO only if real purchasing, sustained use, measured economics, and repeatable market activation support it. Otherwise Olera will preserve and publish the evidence and refine, pivot, or stop the unsupported commercial pathway rather than scale a potential invalidated product or revenue model. Specific post-CRP decision criteria are below:

Measure	Criterion	Source
CareNavigator replication	Verified care or aid establishment reproduced across eight new markets using the same outcome definition as Aim 2	Family confirmation + source records; Task 3.1

Caregiver Staffing paid adoption	Providers purchase and continue using Caregiver Staffing at the Aim 2-informed price	Offer + billing records; Task 3.1
Caregiver Staffing commercial economics	CAC, cost to serve, revenue, and service margin measured from live operations and support a viable provider-paid model	Billing + operating ledger; Task 31
Post-CRP commercialization evidence	CareNavigator evidence supports a credible institutional-buyer proof-of-concept; Caregiver Staffing evidence supports a GO/NO-GO decision for broader commercial expansion	Integrated Aim 2–3 evidence; Task 3.2

Table 3. Aim 3 success criteria, and the source of each measurement.

Timetable. The work is deliberately staged so that downstream exposure occurs only after the preceding gate is met, while independent workstreams proceed in parallel where safe. The governing contingency rule is scope narrows before the calendar compresses: if Aim 1 verification slips, only workflow classes that pass the gate enter Aim 2 and unsupported classes remain human-assisted; if Aim 2 accrual or follow-up slips, Aim 3 does not begin before outcome ascertainment is complete; and if insufficient time remains for both paid-market waves, Wave 1 is completed and analyzed before Wave 2 is opened. This prevents schedule pressure from overriding technical or human-protection gates.

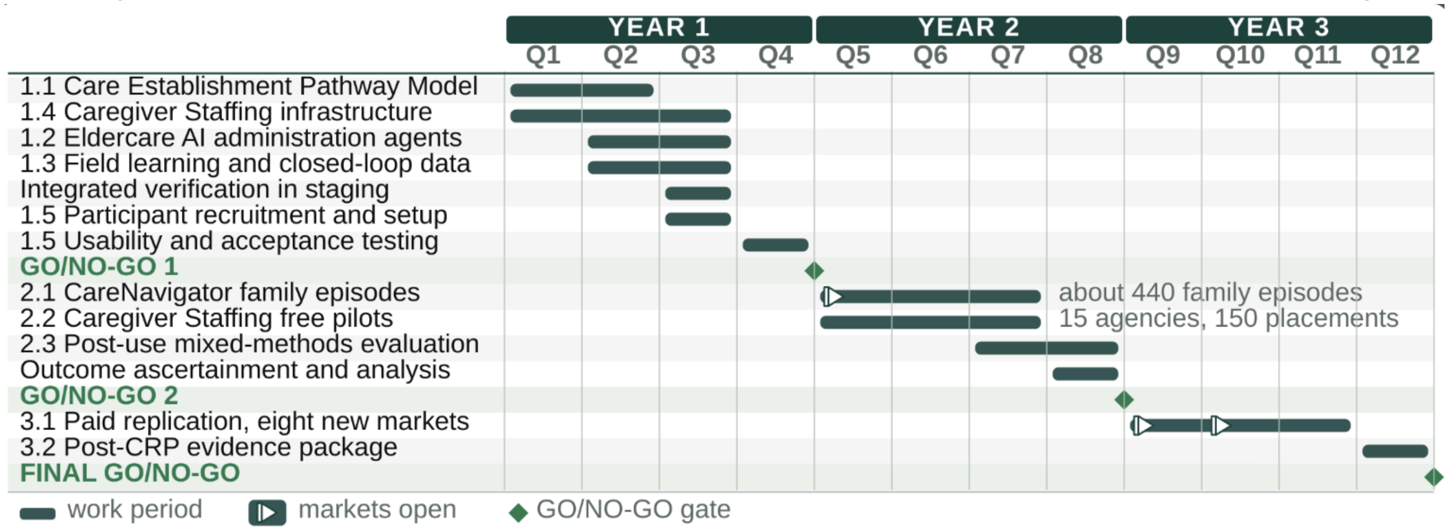

Figure 9. Three-year task-level timetable. GO/NO-GO gates prevent progression to the next level of human or commercial exposure until prespecified criteria are met.

CRP PROGRESS REPORT

What prior funding established. Across NIA SBIR Phase I/II Fast-Track and Phase IIB awards (1R44AG074116) Olera built and evaluated the CareNavigator platform. Olera has also utilized NIH and NSF Innovation-Corps program support to test commercialization risks. Table 6 shows specific de-risking efforts:

Risk	Retired by	Evidence
Technical foundation Could fragmented eldercare data be assembled at national scale?	NIA Phase I to IIB	An expert-curated national database of more than 72,000 aid programs and providers across all fifty states, 578 program guides, and a multi-agent eldercare AI entering production.
User acceptance Would families use and accept the technology?	NIA Phase I to IIB	Four peer-reviewed evaluations with family caregivers: usability 4.57 of 5; acceptance 5.83 of 7 after four weeks (n = 65); multi-agent version 5.73 of 7 (n = 31).
Family demand Can Olera reach families at meaningful scale?	NIA Phase I to IIB	Organic traffic grew from roughly 50 visits a day in 2023 to more than 500 today, from nearly every county, at near-zero acquisition cost.
Provider participation Will providers engage with Olera?	I-Corps and company funds	More than 200 customer-discovery conversations with owners and operators of senior care businesses; more than 700 providers have since claimed an Olera profiles.

Workforce acquisition Can Olera reach an incremental caregiver labor pool?	Company funds	A student caregiver pilot drew more than 900 applicants and placed 25 into provider jobs, establishing a workforce channel Olera can reach directly.
Early willingness to pay Will providers exchange real money for the value Olera creates?	Company funds	Four providers trialed Caregiver Staffing and three paid, at roughly $275 per placement and used the service across multiple semesters. Delivery was deliberately manual, to learn the workflow before building the infrastructure that automates it.
The risks that remain Can the CareNavigator system execute the care-establishment pathway, and Caregiver Staffing provide new careworkers to the workforce? Can the value of these two products be converted into repeatable commercial economics?	Not yet retired	Aim 1 verifies the technology; Aim 2 measures verified care establishment and stakeholder value across eight free validation markets; Aim 3 tests paid conversion, retention, unit economics, and market-entry replication across eight new paid markets.

Table 6. Commercialization risks retired to date, the evidence that retired each, and the risks the CRP is designed to remove.

Why the remaining work is needed. Olera has shown that the care navigation information can be assembled and modeled, that families will use the technology, and that the company can reach families, providers, and workers through channels it controls. The workforce pilot and the first provider payments show that the staffing pathway can move people into provider jobs and that providers will pay for what Olera produces. Delivering that manually exposed the workflow that now has to be engineered. What none of it establishes is whether the integrated CareNavigator and Caregiver Staffing infrastructure can execute the care-establishment pathway reliably, whether that execution reaches established care and creates stakeholder value in real markets, or whether the resulting value converts into repeatable commercial economics. Those three uncertainties are assigned to Aims 1 to 3.

Team and execution capacity. The proposed work is led by substantially the same research and operating team that built and evaluated the system. Falohun was PI on both prior NIA awards and leads product, technical integration, and the Aim 1 engineering workstream. DuBose, Olera's Chief Research Officer and Co-Investigator on Phase I–IIB, leads research integration, milestone coordination, and cross-aim operations. Fan and Clemson independently lead the human-subjects work, with Ory advising as she has on Phase IIB. Qu coaches the founders Falohun and DuBose on commercial milestones and investor readiness. Olera also enters the CRP with a full-time software engineer, two full-time family/provider operations personnel with more than three years of platform experience, research and marketing support, and at least one additional full-time engineering position planned for CRP Year 1. One existing operations staff member will be designated Market Operations Lead before Aim 2 and will execute the non-founder market-entry test in Aim 3.